

European consensus-based (S2k) Guideline on the Management of Herpes Zoster – Guided by the European Dermatology Forum (EDF) in cooperation with the European Academy of Dermatology and Venereology (EADV)

Part 1: Diagnosis

Clinical Background / Introduction

Herpes zoster (HZ, shingles) and zoster-associated pain (ZAP) result from reactivation of varicella zoster virus (VZV), which remains latent in sensory nerve ganglia following primary infection. Primary VZV infection typically occurs in childhood and manifests as varicella (chickenpox), characterized by a generalized rash. During this acute phase, the virus establishes latency in sensory neurons of the dorsal root ganglia (DRG) along the entire neuroaxis.

Reactivation of VZV occurs when virus-specific cell-mediated immunity declines, commonly due to aging or immunosuppression. This reactivation leads to viral replication within one or more DRG. Virions are then transported antidromically along axons via the microtubular system to the skin. Upon reaching intra-epidermal nerve endings and the perifollicular neural network, the virus replicates in epidermal or infundibular keratinocytes.

Viral replication disrupts normal keratinocyte differentiation, leading to cytopathic changes such as cell ballooning, multinucleated giant cells, syncytia formation, eosinophilic intranuclear inclusions (Cowdry type A), and ultimately apoptosis. These changes clinically manifest as grouped vesicles on an erythematous base, typically in a unilateral, dermatomal distribution.

Diagnosis of Herpes Zoster

Clinical Diagnosis

The diagnosis of HZ is primarily clinical, based on characteristic history and dermatological findings.

- **Prodromal phase:** Patients may experience localized pain, itching, or tingling in a dermatomal distribution days to weeks before rash onset. This can be accompanied by systemic symptoms such as malaise, fever, or headache.
- **Rash presentation:** The hallmark of HZ is a unilateral, dermatomal vesicular rash that does not cross the midline. Lesions typically progress from macules to papules, then to clustered vesicles on an erythematous base, and finally to pustules and crusts over 7–10 days.
- **Pain:** Zoster-associated pain (ZAP) varies in intensity and may precede, accompany, or follow the rash. It can be sharp, burning, or throbbing and is often exacerbated by light touch (allodynia).
- **Atypical presentations:** In immunocompromised individuals or older adults, HZ may present with:
 - Disseminated rash (≥ 20 vesicles outside the primary or adjacent dermatomes)
 - Chronic verrucous or ulcerative lesions
 - Absence of rash (zoster sine herpete – rare and difficult to diagnose)
 - Motor involvement (e.g., facial nerve palsy in Ramsay Hunt syndrome)

Differential Diagnosis

Conditions that may mimic HZ include: - Herpes simplex virus (HSV) infection - Contact dermatitis - Impetigo - Dermatomal erythema multiforme - Bullous pemphigoid - Linear IgA disease - Insect bites - Early stages of erythema multiforme or cellulitis

Laboratory Confirmation

While clinical diagnosis is usually sufficient, laboratory testing is recommended in atypical cases, immunocompromised patients, or when diagnostic uncertainty exists.

Preferred Methods:

- **Polymerase chain reaction (PCR):** - Highly sensitive and specific for detecting VZV DNA - Can be performed on vesicular fluid, crusts, or tissue samples - Allows differentiation between wild-type and vaccine strains - Recommended method for confirming VZV in suspected cases of HZ, especially in atypical presentations or CNS involvement
- **Direct fluorescent antibody (DFA) testing:**
 - Rapid method using vesicle scrapings or biopsy specimens
 - Moderate sensitivity compared to PCR
 - Useful in settings where PCR is not readily available
- **Serology:**
 - Not routinely recommended for acute diagnosis
 - May be helpful in cases of suspected zoster sine herpete or retrospective diagnosis
 - Detection of VZV-specific IgM indicates recent infection/reactivation but has limited sensitivity and specificity
 - Paired serology (acute and convalescent samples) showing a fourfold rise in IgG titer may support diagnosis but is rarely practical in clinical practice

- **Viral culture:**
- Low sensitivity and slow turnaround time
- Not recommended for routine use
- **Histopathology and immunohistochemistry (IHC):**
- Skin biopsy may show characteristic cytopathic effects: acantholysis, multinucleated giant cells, ballooning degeneration, eosinophilic nuclear inclusions
- IHC can confirm presence of VZV antigens in tissue
- Reserved for cases with unusual morphology or suspected systemic involvement

Special Considerations

Ophthalmic Herpes Zoster (HZO)

- Involves the ophthalmic division of the trigeminal nerve (V1)
- Key signs: vesicles on the forehead, scalp, or nose (Hutchinson's sign – indicates nasociliary involvement and higher risk of ocular complications)
- Requires urgent ophthalmologic evaluation due to risk of keratitis, uveitis, glaucoma, and vision loss
- PCR testing of conjunctival or corneal swabs may aid in early diagnosis

Ramsay Hunt Syndrome

- Caused by VZV reactivation in the geniculate ganglion
- Triad: auricular vesicles, facial nerve palsy, and hearing loss or vertigo
- Diagnosis is clinical; PCR of vesicle fluid confirms VZV
- Early treatment improves prognosis for facial nerve recovery

Neurological Involvement

- VZV can cause meningoencephalitis, myelitis, or vasculopathy (e.g., stroke in elderly patients)

- In suspected CNS involvement, PCR analysis of cerebrospinal fluid (CSF) is essential
- CSF findings may include lymphocytic pleocytosis, elevated protein, and positive VZV PCR
- MRI brain/spine may show inflammatory or ischemic lesions

Disseminated Herpes Zoster

- Defined as ≥ 20 vesicles outside the primary or adjacent dermatomes
- Typically seen in immunocompromised individuals
- May progress to visceral involvement (e.g., hepatitis, pneumonitis)
- Requires hospitalization and intravenous antiviral therapy
- PCR from blood or bronchoalveolar lavage may be necessary in systemic disease

Summary of Key Diagnostic Recommendations

- **Clinical diagnosis** of HZ is sufficient in typical cases with classic dermatomal vesicular rash and pain.
- **PCR testing** is the preferred method for laboratory confirmation in atypical, immunocompromised, or complicated cases.
- **DFA** may be used if PCR is unavailable.
- **Serology** has limited utility in acute diagnosis and is not routinely recommended.
- **Biopsy with histopathology and IHC** should be considered in atypical cutaneous lesions or suspected systemic involvement.
- **Ocular involvement** must be ruled out in all cases of HZ involving the V1 dermatome; referral to ophthalmology is mandatory if suspected.
- **Neurological symptoms** (e.g., headache, confusion, focal deficits) warrant evaluation for CNS involvement with CSF analysis and neuroimaging.
- **Disseminated disease** requires prompt recognition and systemic antiviral treatment.

> Note: This guideline emphasizes the importance of timely and accurate diagnosis to initiate appropriate management and prevent complications such as postherpetic neuralgia (PHN), vision loss, or neurological sequelae.